

PROJECT REPORT

MOVIE TICKET BOOKING SYSTEM

Python Programming Project

Submitted as a First Semester Python Project

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1. Introduction

The Movie Ticket Booking System is a small Python project made to practice the basic Python concepts learned in the first semester. The program allows a user to select a movie, choose a show timing, book seats and enter basic customer details. It also generates a Booking ID for the booking.

I selected this project because movie ticket booking is a simple real-life situation where Python concepts like lists, dictionaries, loops, conditions, input and basic calculations can be used together.

2. Problem Statement

The aim of the project is to make a simple movie ticket booking program using Python. The program should display available movies and allow the user to select a movie. After selecting a movie, its available show timings and ticket price are displayed.

The user can then enter the number of seats required. The program calculates the total ticket amount and asks for the customer's name and phone number. After the booking is completed, a Booking ID is generated. The booking can also be viewed using the Booking ID and can be cancelled.

3. Objectives

- To display the available movies.
- To allow the user to select a movie.
- To display show timings for the selected movie.
- To calculate the total ticket price.
- To keep track of available seats.
- To take basic customer details.
- To generate a Booking ID.
- To display booking details.
- To allow cancellation of a booking.
- To return cancelled seats to the available seats.

4. Technologies Used

- Programming Language: Python
- Development environment: Python / Jupyter Notebook
- Data structures used: Lists and Dictionary
- No external database or library is required.

5. Python Concepts Used

- Lists - used to store movies and show timings.
- Nested lists - used to keep movie details together.

- Dictionaries - used to store the booking details.
- Variables - used to store choices, seats, price and customer details.
- for loop - used to display movie and show options.
- if-else statements - used for checking valid choices and booking conditions.
- input() - used to take information from the user.
- String operations - used for customer information and Booking ID.
- Basic calculations - used to calculate the total ticket amount and available seats.

6. Working of the Project

- **Step 1:** The program displays the available movies.
- **Step 2:** The user enters the movie number.
- **Step 3:** The selected movie's genre, timings and ticket price are displayed.
- **Step 4:** The user selects a show timing.
- **Step 5:** The program starts with 40 available seats and asks how many seats the user wants.
- **Step 6:** The total amount is calculated using the number of seats and ticket price.
- **Step 7:** The customer name and phone number are taken.
- **Step 8:** A Booking ID is generated and the booking details are displayed.
- **Step 9:** The user can enter the Booking ID to view the booking.
- **Step 10:** The user can cancel the booking. If cancelled, the booked seats are added back to the available seats.

7. Main Features

- Movie selection
- Show timing selection
- Ticket price display
- Seat booking
- Total amount calculation
- Customer details
- Booking ID generation
- Booking details search
- Booking cancellation
- Returning cancelled seats

8. Project Files

File	Purpose
movie_booking.py	Main Python source code of the movie booking program.
README.md	Contains a short explanation of the project, features and running instructions.
statement.md	Contains the problem statement and main objectives.

9. Sample Working

For example, the user can select a movie from the displayed list and then select one of its available show timings. The program displays the ticket price and asks for the number of seats. If the user books valid seats, the total amount is calculated and the remaining seats are shown.

After entering the customer's name and phone number, the program displays a booking confirmation with the Booking ID, customer name, movie, show, number of seats and total amount.

The program also asks for a Booking ID to view the saved booking. During cancellation, the booking is checked and the booked seats are returned to the available seats when the user confirms cancellation.

10. Testing

I tested the main flow of the program using valid inputs. The program was able to display movies, accept a movie choice, display timings, accept the seat count, calculate the amount, take customer details, generate a Booking ID, display booking details and cancel the booking.

The cancellation part also updates the available seats by adding the cancelled seats back.

11. Limitations

- The project is a basic version and does not use a database.
- It mainly handles one booking at a time.
- The program expects the user to enter the correct type of input.
- It does not have a graphical user interface.
- The booking information is not permanently saved after the program is closed.

12. Conclusion

This project helped me understand how basic Python concepts can be used together to make a practical program. I used lists, dictionaries, loops, conditions, user input and basic calculations in the project.

The Movie Ticket Booking System is a simple first-semester project, but it helped me get more practice with Python programming and understand how a small real-life problem can be converted into a program.